

Component Diagram

It is a structural UML diagram that shows how a system is wired together using independent pieces of software & API/Service they use to talk to each other.

Structure of Component Diagram:

1) Component:

It is a modular, encapsulated & replaceable part of a system.

4 types of things can be a component:

1) Deployables (executables):

Any software that a user actually downloads, installs or opens in web.

Keywords: Mobile app, Desktop Client, etc.

BHATTI

• Basically front end components.

2) Storage :

Anywhere data is saved is made into a component.

Keywords: Data base, Repository, Cloud Storage

• Data component.

3) 3rd - party services:

Highlight any system that your system didn't build but needs to use.

Keywords: Payment gateway (PayPal/Payoneer), SMS service, email service etc.

• External Component.

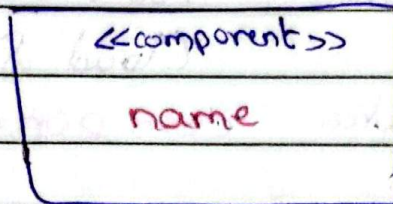
4) Engines (core logic / Micro services):

If a system is complex, it might have multiple internal services.

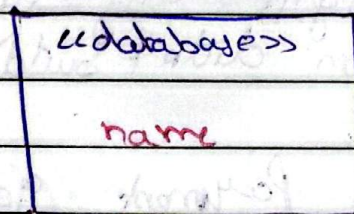
Keywords: API, system, Module, Engine etc.

• Backend components.

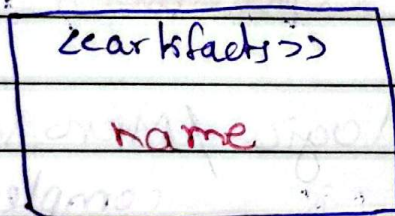
Representation:



• Generally



• Specifically for DB

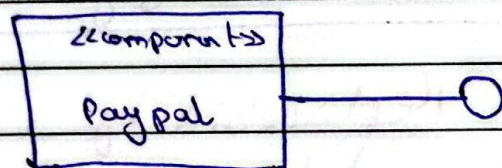


- This is the specification of physical piece of info such as
• exe, dll or
• Java class
file.

• «database» is also a special type of «artifact»

2) Lollipop (Provided interface):

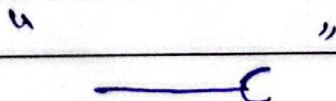
It is a service or function that a component offers to the rest of system.



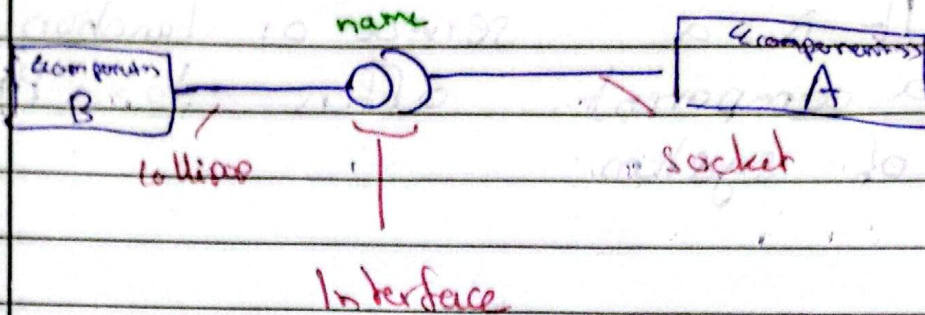
This means paypal is offering a service (payment).

3) Socket (Required interface):

This shows that a component ~~is~~ needs a service, function or data from someone outside of it.



The ball & Socket Join:



we can
also
name
interface)

This shows A needs B's services.

Why is this Join so significant:

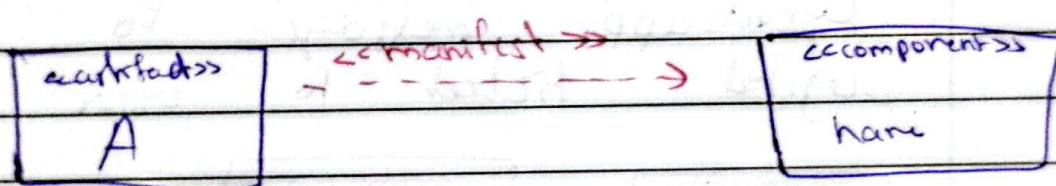
It shows **Replaceability**, makes our design modular. We can easily change Any service just by connecting another Lollipop.

It also shows use of **Microservices**.

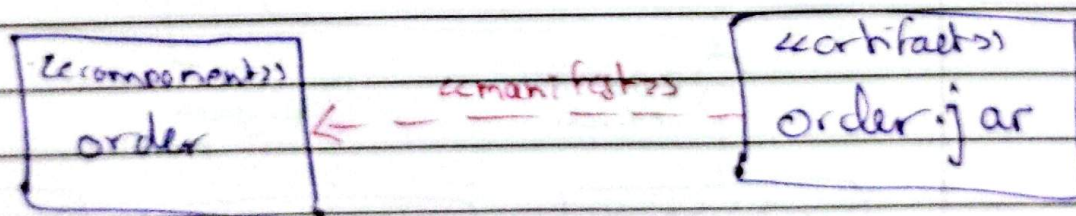
4) Manifestation Arrow:

An artifact is said to manifest model elements that are used to construct it.

basically if an artifact is created by some component we will have a manifestation arrow from artifact to component.



~~example~~
example



Example:

Scenario: A cinema needs a Ticket Booking System. Customers can buy tickets using either web app or Mobile App. When customer buys, central Booking API processes request. API saves booking details to Central Movie DB. To charge customer Booking API sends the credit card details to PayPal. Once paid API uses ~~text~~ whatsapp message to send the digital ticket to user.

